The δ-commutator lemma is a butterfly

The automorphism crossed module of the inventory gerbe

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Abstract

The obstruction to reconciling fungible inventory across a supply chain is, following Giraud, a *tower*: a nonabelian band $\beta \in H^1(N(\mathcal{C}); \mathrm{Out}K)$ in the relabelling group $K = \mathrm{Aut}F$ must vanish before an abelian inventory class is even defined, and on the torus these two stages turn out to couple through a connecting map given exactly by a commutator of automorphism lifts, $\delta(\rho) = [[A, B]]$. This note re-reads that result in the language of crossed modules and butterflies. The whole tower is a *single* crossed module $\partial\colon K \to \mathrm{Aut}K$: its cokernel $\mathrm{Out}K$ is the band's target, its kernel $Z(K)$ is the inventory coefficient, and its image $\mathrm{Inn}K$ is the hinge the two Giraud extensions share. A band is a map into the π_0 of the associated 2-group; it lifts strictly iff it extends to a strict morphism of crossed modules, and it *always* extends to a weak morphism, presented — following Aldrovandi and Noohi — by a *butterfly*. The connecting map is the class of the butterfly's waist extension, and the commutator $[A, B]$ is literally the commutator of two lifts computed inside the waist group. The bar–Koszul comparison map that pins the class to this commutator is the butterfly's concrete cochain model. No result is new; the aim is a picture in which the shared hinge, and the commutator formula, become inevitable.

1 Introduction

The result being re-read. A supply chain is a small category $\mathcal{C}$; inventory is a datum glued along its morphisms. When local stock charts are comparable only *up to* a relabelling of states — up to the automorphism group $K = \mathrm{Aut}F$ of the stock fibre F — the gluing datum is an $\mathrm{Aut}F$-gerbe over the nerve $N(\mathcal{C})$, and the obstruction to a single globally coherent inventory is, after Giraud [5, 2], not one class but a staged tower. The relabelling group sits in two group extensions that share their middle term $\mathrm{Inn}K$:

$$\text{(I)}\quad 1 \to \mathrm{Inn}K \to \mathrm{Aut}K \xrightarrow{q} \mathrm{Out}K \to 1, \qquad \text{(II)}\quad 1 \to Z(K) \to K \to \mathrm{Inn}K \to 1. \tag{1}$$

The *band* $\beta \in H^1(N(\mathcal{C}); \mathrm{Out}K)$ is read off (I); it must vanish before the abelian *inventory class* $[\mathfrak{g}] \in H^2(N(\mathcal{C}); Z(K))$ of (II) is even defined. The companion proof note [6] showed that over the torus $T^2 \simeq N(B\mathbb{Z}^2)$ these two stages no longer clear independently: writing a band as a

*Kodamai. This is an *expository* companion to the proof note "The inventory gerbe couples on the torus" (content commit `e306ae5`, cited below as [6]); it introduces no new theorem and leaves every grade of that note unchanged. The present note lives in the author's `work-in-progress` corpus, content commit `0d73790`. Prepared for the Kodamai AI-Mathematician grant (Theory pillar) and for Neil Ghani's Strathclyde mini-course *Tangent Categories and Containers*, 13–16 October 2026.

homomorphism $\rho\colon \mathbb{Z}^2 \to \mathrm{Out}K$ (equivalently a *commuting pair* $\rho(s), \rho(t)$, since $\mathbb{Z}^2$ is abelian), the connecting map from band to strictification is given exactly by the commutator of any two lifts,

$$\delta(\rho) = [\,[A, B]\,] \in H^2(T^2; \mathrm{Inn}K), \qquad A, B \in \mathrm{Aut}K,\ q(A) = \rho(s),\ q(B) = \rho(t). \tag{2}$$

That lemma was proved by a deliberately elementary Schreier-style computation. This note does not reprove it. It re-reads it.

The gap this note fills. The proof of (2) is clean but its *shape* is hidden. Three facts sit side by side in the computation without an evident reason to be together: that the two Giraud extensions (1) share the middle term $\mathrm{Inn}K$; that the connecting map is a commutator; and that the right coefficient for the strictification stage is $\mathrm{Inn}K$ while the right coefficient for the inventory stage is $Z(K)$. Why *these* groups, in *this* arrangement? The elementary proof answers "because the arithmetic works out". We want a reason one can see.

What this note does. The reason is a single crossed module. Recall that a crossed module is a homomorphism $\partial\colon C \to G$ with a G-action on C satisfying two compatibilities (§2); it is the algebraic incarnation of a 2-group, and it has two homotopy groups $\pi_0 = \mathrm{coker}\partial$ and $\pi_1 = \ker\partial$. The group K carries a canonical one, the *automorphism crossed module*

$$\mathsf{AUT}(K)\colon \qquad \partial = \mathrm{conj}\colon K \longrightarrow \mathrm{Aut}K, \qquad k \mapsto c_k \text{ (conjugation by } k),$$

and its invariants are exactly the players in (1):

$$\pi_0\mathsf{AUT}(K) = \mathrm{Out}K, \qquad \pi_1\mathsf{AUT}(K) = Z(K), \qquad \mathrm{im}\,\partial = \mathrm{Inn}K.$$

The first astonishment (§2) is that the two Giraud extensions are *not two facts*: extension (I) is the cokernel extension of ∂ and (II) is its kernel extension, so they are the two edges of one crossed module, necessarily welded at the image $\mathrm{Inn}K$. The shared hinge is forced.

A band $\rho\colon \mathbb{Z}^2 \to \mathrm{Out}K$ is then a map into $\pi_0\mathsf{AUT}(K)$ (§3). It lifts to a strict homomorphism $\mathbb{Z}^2 \to \mathrm{Aut}K$ iff it extends to a *strict* morphism of crossed modules — and generically it does not. But the honest notion of morphism between crossed modules, the one that matches maps of the underlying 2-group stacks, is the *weak* morphism, and every band extends to one. Following Aldrovandi and Noohi [1], a weak morphism is presented by a *butterfly*: a span of groups whose central *waist* is a group extension (§4). The second astonishment (§§5–6) is that the proof's pullback extension E_ρ *is* the waist of this butterfly (up to the centre), so that

$$\delta(\rho) = \text{the class of the waist extension},$$

and the commutator $[A, B]$ of (2) is literally the commutator of two lifts computed *inside the waist group* and pushed to $\mathrm{Inn}K$ by ∂. The bar–Koszul comparison map $\varphi_2(e_{st}) = [s\,|\,t] - [t\,|\,s]$ that the proof uses to pin the class is the concrete cochain shadow of reading this commutator off the waist. Both Giraud stages are instances of one construction, unified by $\mathsf{AUT}(K)$ (§7).

Status. This is exposition. The underlying lemma, witnesses, and sharp line are proved and computed in [6]; the crossed-module and butterfly translations below are routine unwindings of definitions, stated as observations, not theorems. The one external input is the butterfly formalism of [1], whose bitorsor ancestry is Breen's [2]; the classical $H^2 \leftrightarrow$ extensions dictionary is Brown [3] and Eilenberg–Mac Lane [4]. The intended reader is comfortable with group cohomology and short exact sequences but has not met 2-groups or butterflies; §2 starts from the definition.

Outline. Section 2 introduces crossed modules and the automorphism crossed module, and locates the two Giraud extensions inside it. Section 3 reads the band as a map into π_0 and the strict-lifting problem as a strict-morphism-extension problem. Section 4 introduces weak morphisms and butterflies, with $\mathsf{AUT}(K)$ as the running instance. Section 5 identifies the connecting map with the waist class, and Section 6 identifies the commutator formula with a computation in the waist and its bar–Koszul shadow. Section 7 returns to the shared hinge and both stages, and names what the language buys.

2 The automorphism crossed module

2.1 Crossed modules, gently

Definition 1. A *crossed module* is a group homomorphism $\partial\colon C \to G$ together with a left action of G on C by automorphisms, written $g \cdot c$, such that

(E) $\partial(g \cdot c) = g\,\partial(c)\,g^{-1}$ (*equivariance*), and

(P) $\partial(c) \cdot c' = c\,c'\,c^{-1}$ (*Peiffer identity*),

for all $g \in G$, $c, c' \in C$. Its *homotopy groups* are

$$\pi_0(\partial) = \operatorname{coker}\partial = G/\operatorname{im}\partial, \qquad \pi_1(\partial) = \ker\partial.$$

A crossed module is the data of a strict 2-group (equivalently a strict categorical group): G is the group of objects, $G \ltimes C$ the group of arrows, source and target $s(g) = g$, $t(g, c) = \partial(c)\,g$. The group π_0 is the set of isomorphism classes of objects — the "components" — and π_1 is the group of automorphisms of the identity object. A reader who prefers topology may keep in mind the classifying space $B\mathsf{AUT}$, a connected space with $\pi_1 = \pi_0$ and $\pi_2 = \pi_1$ and nothing higher; crossed modules are exactly the algebra of such homotopy 2-types. We will not need the 2-group explicitly until §3; for now $\partial\colon C \to G$ with (E) and (P) is all that is in play.

The Peiffer identity (P) deserves a word, because it is the condition that makes the two homotopy groups behave. It forces $\ker\partial$ to be central in C and to be acted on trivially by $\operatorname{im}\partial$: if $\partial(c) = 1$ then $c\,c'c^{-1} = \partial(c) \cdot c' = c'$ for all c', so $\ker\partial \subseteq Z(C)$, and π_1 is abelian. This is not a technicality we will skip past — it is precisely why the inventory coefficient will come out abelian.

2.2 The automorphism crossed module

Proposition 2. *Let K be any group. The conjugation map*

$$\partial = \mathrm{conj}\colon K \longrightarrow \mathrm{Aut}K, \qquad k \mapsto c_k \quad (c_k(x) = kxk^{-1}),$$

with $\mathrm{Aut}K$ acting on K by evaluation ($\varphi \cdot k := \varphi(k)$), is a crossed module.

Proof. Both axioms are immediate and worth seeing once. For (E), $\partial(\varphi \cdot k) = c_{\varphi(k)}$, while $\varphi\,\partial(k)\,\varphi^{-1} = \varphi\,c_k\,\varphi^{-1} = c_{\varphi(k)}$ since $\varphi(kxk^{-1}) = \varphi(k)\varphi(x)\varphi(k)^{-1}$. For (P), $\partial(k) \cdot k' = c_k(k') = kk'k^{-1}$. □

This $\mathsf{AUT}(K) = [\,K \xrightarrow{\mathrm{conj}} \mathrm{Aut}K\,]$ is the *automorphism crossed module* of K. Its invariants read off directly, and they are the whole cast of (1):

$$\pi_0\mathsf{AUT}(K) = \mathrm{Aut}K/\mathrm{Inn}K = \mathrm{Out}K, \qquad \pi_1\mathsf{AUT}(K) = \ker(\mathrm{conj}) = Z(K), \qquad \operatorname{im}\partial = \mathrm{Inn}K. \quad (3)$$

(That $\ker\mathrm{conj} = Z(K)$ is the statement $c_k = \mathrm{id} \iff k \in Z(K)$; that π_1 is abelian is the Peiffer consequence above, and here it is simply that $Z(K)$ is abelian.)

2.3 The first astonishment: the tower is one crossed module

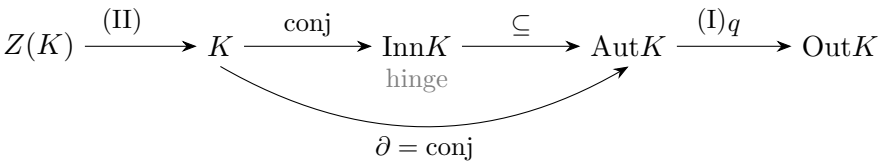
A single homomorphism $\partial\colon C \to G$ breaks into two short exact sequences through its image. One always has

$$1 \to \ker\partial \to C \to \operatorname{im}\partial \to 1 \qquad \text{and} \qquad 1 \to \operatorname{im}\partial \to G \to \operatorname{coker}\partial \to 1,$$

glued along $\operatorname{im}\partial$. For $\partial = \mathrm{conj}\colon K \to \mathrm{Aut}K$ these are, by (3), *exactly* the two Giraud extensions:

$$\underbrace{1 \to Z(K) \to K \to \mathrm{Inn}K \to 1}_{\text{(II): kernel extension}} \qquad \text{and} \qquad \underbrace{1 \to \mathrm{Inn}K \to \mathrm{Aut}K \xrightarrow{q} \mathrm{Out}K \to 1}_{\text{(I): cokernel extension}}. \tag{4}$$

Observation 3. The two extensions (1) of the inventory gerbe are the kernel extension and the cokernel extension of the single crossed module $\mathsf{AUT}(K)$. They share the hinge $\mathrm{Inn}K$ because $\mathrm{Inn}K = \operatorname{im}\partial$ is at once the cokernel of $\mathrm{conj}\colon K \to \mathrm{Inn}K$ (the quotient in (II)) and the kernel of $q\colon \mathrm{Aut}K \to \mathrm{Out}K$ (the normal subgroup in (I)). No coincidence is needed: $\operatorname{im}\partial$ is the unique group through which any homomorphism factors.



The crossed module $\partial\colon K \to \mathrm{Aut}K$ is the lower arc; it factors as the surjection $K \twoheadrightarrow \mathrm{Inn}K$ (giving (II) on the left) followed by the inclusion $\mathrm{Inn}K \hookrightarrow \mathrm{Aut}K$ (giving (I) on the right). Reading the diagram once makes the staged structure of the gerbe obstruction visible as a single object rather than a pair of accidents: the band lives on the right edge, the inventory class on the left edge, and the kernel $Z(K)$ and cokernel $\mathrm{Out}K$ are the two homotopy groups of the 2-group that the whole thing is.

3 The band as a map into π_0

3.1 Bands

Fix throughout a base $T^2 \simeq N(B\mathbb{Z}^2)$: two commuting reversible operations. A band on T^2 is a class in $H^1(T^2; \mathrm{Out}K)$, which for the free base $B\mathbb{Z}^2$ is exactly a homomorphism

$$\rho\colon \mathbb{Z}^2 \longrightarrow \mathrm{Out}K,$$

and, because $\mathbb{Z}^2 = \langle s, t \mid [s,t]\rangle$ is abelian, exactly a *commuting pair* $\rho(s), \rho(t) \in \mathrm{Out}K$. Under (3), $\mathrm{Out}K = \pi_0\mathsf{AUT}(K)$: a band is a map from $\mathbb{Z}^2$ *into the components of the automorphism* 2*-group.* The strictification question of the Giraud tower — can the band be realised by an honest, inner-free transport? — is the question of *lifting* ρ *through the* 2*-group.*

3.2 Strict morphisms are too rigid

The flat-footed way to lift ρ is to ask for a homomorphism $\tilde\rho\colon \mathbb{Z}^2 \to \mathrm{Aut}K$ with $q \circ \tilde\rho = \rho$: a *strict lift.* In crossed-module terms this is a strict morphism. Regard $\mathbb{Z}^2$ as the *discrete* crossed module $[\,1 \to \mathbb{Z}^2\,]$ — trivial in degree one, so $\pi_0 = \mathbb{Z}^2$ and $\pi_1 = 0$. A strict morphism $[\,1 \to \mathbb{Z}^2\,] \to [\,K \to$

$\mathrm{Aut}K\,]$ is, by definition, a pair of homomorphisms (one in each degree) commuting with the two ∂'s and the actions; in degree one the only map out of the trivial group is trivial, so a strict morphism is *nothing but* a homomorphism $\mathbb{Z}^2 \to \mathrm{Aut}K$ lifting ρ on π_0.

Observation 4. A strict morphism of crossed modules $[\,1 \to \mathbb{Z}^2\,] \to \mathsf{AUT}(K)$ inducing ρ on π_0 is the same datum as a strict lift $\tilde{\rho}\colon \mathbb{Z}^2 \to \mathrm{Aut}K$ of the band, i.e. a pair of *commuting* lifts $A = \tilde{\rho}(s)$, $B = \tilde{\rho}(t)$.

And here is the difficulty the whole subject is about: strict lifts usually do not exist. Two outer classes may commute in $\mathrm{Out}K$ while *no* choice of representatives commutes in $\mathrm{Aut}K$. The band ρ remembers only the components; whether it lifts remembers the 2-group. The strict morphism is the wrong tool because it demands the lift on the nose. The next section supplies the right tool, which demands only that the lift exist *up to coherent inner slack* — and then measures the slack.

4 Weak morphisms and butterflies

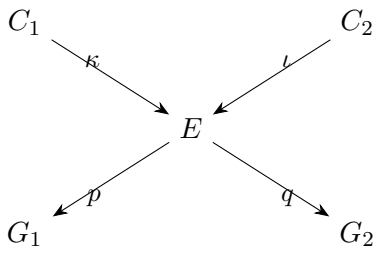
4.1 Why weak morphisms

Crossed modules present 2-groups, and the honest morphisms of 2-groups — the ones that correspond to maps of classifying spaces, or of the associated stacks — are *weak*: they need respect the structure only up to coherent isomorphism. Inverting the weak equivalences of crossed modules (the morphisms that are isomorphisms on both π_0 and π_1) turns the strict morphisms into a larger class of fractions. Aldrovandi and Noohi [1] gave these fractions a concrete, diagram-level description — the *butterfly* — so that one never has to leave the category of groups to compute with weak morphisms. We state the definition in the generality we use and immediately specialise.

Definition 5 (Aldrovandi–Noohi [1])**.** Let $[\,C_1 \xrightarrow{\partial_1} G_1\,]$ and $[\,C_2 \xrightarrow{\partial_2} G_2\,]$ be crossed modules. A *butterfly* between them is a group E together with four homomorphisms

$$C_1 \xrightarrow{\kappa} E \xleftarrow{\iota} C_2, \qquad G_1 \xleftarrow{p} E \xrightarrow{q} G_2,$$

arranged as the two diagonals of



such that

(B1) the NE–SW diagonal $1 \to C_2 \xrightarrow{\iota} E \xrightarrow{p} G_1 \to 1$ is *short exact*;

(B2) the NW–SE diagonal $C_1 \xrightarrow{\kappa} E \xrightarrow{q} G_2$ is a *complex* ($q\kappa = 1$);

(B3) $p\,\kappa = \partial_1$ and $q\,\iota = \partial_2$; and

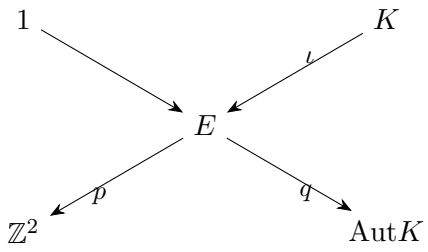
(B4) κ and ι are equivariant for the conjugation action of E on itself (pulled back from the G_i-actions along p and q).

A butterfly represents a weak morphism $[\,C_1 \to G_1\,] \to [\,C_2 \to G_2\,]$, the exact leg (B1) being the weak equivalence one inverts: the span is $G_1 \xleftarrow{p} E \xrightarrow{q} G_2$ with p invertible up to C_2.

The picture earns its name: the two top wings are the degree-one groups mapping *into* the central body E, the two bottom wings are the degree-zero groups E projects *onto*, and the whole is pinched at the waist E. The essential content, for us, is axiom (B1): the body E is an *extension* — of the source's degree-zero group by the target's degree-one group. This is where a cohomology class will live.

4.2 The running butterfly

Specialise to source $[\,1 \to \mathbb{Z}^2\,]$ and target $\mathsf{AUT}(K) = [\,K \xrightarrow{\text{conj}} \mathrm{Aut}K\,]$. Then $C_1 = 1$, $G_1 = \mathbb{Z}^2$, $C_2 = K$, $G_2 = \mathrm{Aut}K$, and a butterfly is a group E with



whose only nontrivial data are the *waist extension* and the map q:

$$\text{(B1)}\quad 1 \to K \xrightarrow{\iota} E \xrightarrow{p} \mathbb{Z}^2 \to 1, \qquad \text{(B3)}\quad q\colon E \to \mathrm{Aut}K,\quad q \circ \iota = \mathrm{conj}. \tag{5}$$

Unwound: a butterfly here is a group E extending $\mathbb{Z}^2$ by the whole of K, together with a homomorphism $q\colon E \to \mathrm{Aut}K$ that restricts on the normal copy of K to conjugation. The weak morphism it carries induces on π_0 the map $\mathbb{Z}^2 \to \mathrm{Out}K$ sending $\gamma \in \mathbb{Z}^2$ to the class of $q(\hat{\gamma})$, where $\hat{\gamma} \in E$ is any element with $p(\hat{\gamma}) = \gamma$ (well-defined modulo $q(K) = \mathrm{Inn}K$). That induced map is the band:

Observation 6. A butterfly E as in (5) induces the band $\rho\colon \mathbb{Z}^2 \to \mathrm{Out}K$ on π_0. A homomorphic section $\tilde{\rho}\colon \mathbb{Z}^2 \to E$ of the waist (if one exists) yields a strict lift $q\tilde{\rho}\colon \mathbb{Z}^2 \to \mathrm{Aut}K$ of ρ — a pair of commuting lifts, as in Observation 4 — so that the weak morphism carried by E becomes isomorphic to a strict morphism.

So the passage from "strict lift" to "butterfly" is the passage from "the band lifts to a homomorphism" to "the band lifts only into an *extension*". The obstruction to going back is a cohomology class, and that class is the connecting map, measured on the quotient E_ρ of the waist (§5).

5 The connecting map is the waist

The proof note [6] forms, for a band ρ, the pullback extension

$$E_\rho := \mathrm{Aut}K \times_{\mathrm{Out}K} \mathbb{Z}^2 = \{(a,\gamma) \in \mathrm{Aut}K \times \mathbb{Z}^2 : q(a) = \rho(\gamma)\}, \qquad 1 \to \mathrm{Inn}K \to E_\rho \xrightarrow{\bar{p}} \mathbb{Z}^2 \to 1,$$

and *defines* $\delta(\rho) := [E_\rho] \in H^2(T^2; \mathrm{Inn}K)$. This is the waist of the running butterfly, pushed down past the centre.

Proposition 7. *Let E be a butterfly as in (5) inducing ρ. The map $(q,p)\colon E \to \mathrm{Aut}K \times \mathbb{Z}^2$ lands in E_ρ and induces an isomorphism*

$$E/\,\iota(Z(K)) \xrightarrow{\sim} E_\rho,$$

carrying the waist extension (5) of $\mathbb{Z}^2$ by K onto the pullback extension of $\mathbb{Z}^2$ by $\mathrm{Inn}K$ along the quotient $K \twoheadrightarrow \mathrm{Inn}K$. Consequently $\delta(\rho) = [E_\rho]$ is the image of the waist class under $K \twoheadrightarrow \mathrm{Inn}K$.

Proof. For $e \in E$ one has $q(e) \in \mathrm{Aut}K$ and $p(e) \in \mathbb{Z}^2$ with $q(e) \bmod \mathrm{Inn}K = \rho(p(e))$ by Observation 6; hence $(q(e), p(e)) \in E_\rho$, so (q,p) is a homomorphism $E \to E_\rho$. It is surjective because p is surjective and $q|_K = \mathrm{conj}$ surjects K onto the fibre $\mathrm{Inn}K$ of E_ρ over 0. Its kernel is $\{e : p(e) = 0,\ q(e) = \mathrm{id}\}$; by (B1), $p(e) = 0$ means $e = \iota(k)$ for a unique $k \in K$, and then $q(e) = \mathrm{conj}(k) = \mathrm{id}$ means $k \in \ker \mathrm{conj} = Z(K)$. So $\ker(q,p) = \iota(Z(K))$, a normal subgroup (being a kernel), and the stated isomorphism is the first isomorphism theorem. The extension statement is the commuting square

$$\begin{array}{ccc} K & \longrightarrow & E \\ \downarrow & & \downarrow \\ \mathrm{Inn}K & \longrightarrow & E_\rho \end{array}$$

with vertical maps the quotients by $Z(K)$, exhibiting E_ρ as the pushout of (5) along $K \twoheadrightarrow \mathrm{Inn}K$. □

Observation 8. Under Proposition 7, the three vanishing conditions line up:

$$\delta(\rho) = 0 \overset{[3,\,\mathrm{IV}.3]}{\iff} E_\rho \text{ splits} \iff \rho \text{ has a strict lift}$$
$$\iff \text{the weak morphism carried by the butterfly is isomorphic to a strict one.}$$

That is the whole conceptual content of the connecting map, now visible: $\delta(\rho)$ is the obstruction to the waist of the butterfly being split, equivalently to the band's weak lift being strictifiable. The proof note establishes the equivalence $\delta(\rho) = 0 \iff$ (commuting lifts) by the classical extension dictionary [3, 4]; here the same equivalence is read off the butterfly. What remains — the *value* of the class when it does not vanish — is a commutator, and it too lives in the waist.

6 The commutator is a commutator in the waist

6.1 The commutator lives upstairs

Choose lifts of the $\mathbb{Z}^2$-generators into the waist: $\hat{s}, \hat{t} \in E$ with $p(\hat{s}) = s$, $p(\hat{t}) = t$. Their images $A := q(\hat{s})$, $B := q(\hat{t}) \in \mathrm{Aut}K$ are lifts of $\rho(s), \rho(t)$ in the sense of (2). Because s and t commute in $\mathbb{Z}^2$, the commutator $[\hat{s}, \hat{t}]$ lies in $\ker p = \iota(K)$, so it *is* an element of K:

$$[\hat{s}, \hat{t}] = \iota(w_0), \qquad w_0 \in K. \tag{6}$$

Apply q and use $q\iota = \mathrm{conj}$:

$$[A, B] = [q(\hat{s}), q(\hat{t})] = q([\hat{s}, \hat{t}]) = q(\iota(w_0)) = \mathrm{conj}(w_0) = c_{w_0} \in \mathrm{Inn}K. \tag{7}$$

Observation 9. The $\mathrm{Inn}K$-valued commutator $w = [A, B]$ of (2) is the image under $\partial = \mathrm{conj}$ of a genuine K-valued commutator $w_0 = [\hat{s}, \hat{t}]$ measured *inside the butterfly's waist*. The outer-level "$[A, B]$ fails to vanish" is the shadow of the honest group commutator of two chosen lifts failing to be central.

This is the cleanest way to see why a commutator appears at all. The band records a pair of commuting *outer* classes; lifting it into the waist chooses honest group elements $\hat{s}, \hat{t}$; they need not commute, and *their commutator is the whole obstruction*, because it is exactly the element of $K \subseteq E$ one picks up on traversing the torus's defining loop $sts^{-1}t^{-1}$.

6.2 The bar–Koszul shadow

To turn (7) into a cohomology class in the correct coordinates, the proof note reads the waist extension through a factor set and compares resolutions. We recall the mechanism only to the depth that shows it is the cochain shadow of Observation 9; the computation itself is [6, §3].

Over $R = \mathbb{Z}[\mathbb{Z}^2] = \mathbb{Z}[s^{\pm1}, t^{\pm1}]$ there are two free resolutions of the trivial module $\mathbb{Z}$: the small *Koszul* resolution, whose top differential encodes the single torus relator and which computes

$$H^2(T^2; M) = M\big/\big((\sigma-1)M + (\tau-1)M\big) = M_{\mathbb{Z}^2} \qquad (\sigma, \tau = \text{the } \mathbb{Z}^2\text{-action}),$$

and the large *bar* resolution, in which a factor set of an extension is literally a 2-cocycle $f(g,h)$. The unique-up-to-homotopy comparison map $\varphi\colon K_\bullet \to B_\bullet$ sends the fundamental 2-cell to the antisymmetric expression

$$\varphi_2(e_{st}) = [\,s \,|\, t\,] - [\,t \,|\, s\,], \tag{8}$$

so that evaluation of any bar 2-cocycle f on the generator of $H^2(T^2; M)$ is $\varphi^*[f] = [\,f(s,t) - f(t,s)\,] \in M_{\mathbb{Z}^2}$. Expression (8) is the algebraic shadow of the loop $sts^{-1}t^{-1}$: the two orderings of the cell, with opposite signs.

Now take $M = \mathrm{Inn}K$ and the factor set of the waist extension E_ρ for the section $u(s^i t^j) = \hat{s}^{\,i}\hat{t}^{\,j}$ (the same lifts as above, pushed to $\mathrm{Inn}K$). One computes, using $st = ts$,

$$f(s,t) = u(s)u(t)u(st)^{-1} = 1, \qquad f(t,s) = u(t)u(s)u(ts)^{-1} = \hat{t}\hat{s}(\hat{s}\hat{t})^{-1} = [\hat{s},\hat{t}]^{-1} = w^{-1}$$

in $\mathrm{Inn}K$. Hence, by (8),

$$\delta(\rho) = \varphi^*[f] = [\,f(s,t) - f(t,s)\,] = [\,0 - (-w)\,] = [\,w\,] = [\,[A,B]\,] \in \mathrm{Inn}K_{\mathbb{Z}^2}. \tag{9}$$

The antisymmetric $[s|t] - [t|s]$ of (8) is nothing but the two orderings $f(s,t)$ and $f(t,s)$, whose quotient is the commutator; the comparison map is the bookkeeping that makes "evaluate on the torus 2-cell" mean "take the commutator of the lifts in the waist". The class $[w]$ is independent of the lifts because changing $\hat{s}, \hat{t}$ by elements of $\ker p = K$ changes w by a coboundary (the lift-independence computation of [6]); in the butterfly this is the statement that different section-choices give isomorphic waist extensions.

Remark 10 (trivial action and the clean slogan)**.** When $\sigma = \tau = \mathrm{id}$ — the case of every explicit witness in [6], where the lifts act trivially on $\mathrm{Inn}K$ — the coboundaries vanish, $\mathrm{Inn}K_{\mathbb{Z}^2} = \mathrm{Inn}K$, and (9) reduces to the bare slogan $\delta(\rho) = [A,B] \neq 0$. The coinvariant quotient $M_{\mathbb{Z}^2}$ is the correction the general lemma carries for a nontrivial action; it is the π_0-coinvariants of π_1-with-a-twist, exactly what a 2-group hands to H^2.

7 The shared hinge, and both stages at once

We treated the band-to-strictification stage, extension (I), with the butterfly into $\mathsf{AUT}(K) = [\,K \to \mathrm{Aut}K\,]$. The inventory stage, extension (II), is the same picture one edge to the left. The general lemma of [6] applies to *any* abelian-kernel extension $1 \to M \to \mathcal{E} \to Q \to 1$ and band $\mathbb{Z}^2 \to Q$; the two Giraud stages are its two instances,

$$\text{(I)}\quad (\mathcal{E}, Q, M) = (\mathrm{Aut}K, \mathrm{Out}K, \mathrm{Inn}K), \qquad \text{(II)}\quad (\mathcal{E}, Q, M) = (K, \mathrm{Inn}K, Z(K)),$$

and both are edges of the one crossed module $\mathsf{AUT}(K)$ by Observation 3. The kernel edge (II) has coefficient $Z(K) = \pi_1$, abelian for *every* K by the Peiffer identity; the cokernel edge (I) has coefficient $\mathrm{Inn}K$, abelian exactly when K has nilpotency class ≤ 2 — the standing hypothesis of [6], now identified as the condition that the *image* of ∂, not merely its kernel, be abelian.

Observation 11. The hinge $\mathrm{Inn}K$ carries both stages: as the *quotient* in (II) it is the target of the inventory band, and as the *kernel* in (I) it is the coefficient of the strictification class. The coupling of the two stages on T^2 — a nonzero band forcing a nonzero class — is therefore a statement about a single crossed module failing to be "split on both edges" over $\mathbb{Z}^2$, and the common mechanism, a commutator of lifts in a waist, is the same on each edge.

What the language buys. The butterfly reformulation proves nothing the Schreier computation did not, but it makes three things structural rather than incidental. First, the shared hinge is forced (Observation 3): one cannot have the two Giraud extensions without their sharing $\mathrm{Inn}K$, because they are the two factorisation-edges of one map. Second, the commutator is forced (Observation 9): the obstruction is a commutator because it is *the* element of the fibre one collects traversing the torus relator in the waist group, not a feature of the particular group K. Third, the coefficients are forced: $\pi_1 = Z(K)$ is abelian by Peiffer, and the class-≤ 2 hypothesis is exactly $\mathrm{im}\,\partial$ abelian — the hypothesis is read off the crossed module, not imposed by hand.

What it suggests. Butterflies compose, and composition of butterflies is composition of weak morphisms of 2-groups [1]. Two welds along a reconvergent logistics network compose as a Zappa–Szép-type overlay; whether that composite weld's obstruction is the composite of the butterflies' waist classes is a clean question the present language poses but does not settle, and it is the natural next node. A second is the passage beyond class ≤ 2: dropping that hypothesis makes $\mathrm{im}\,\partial = \mathrm{Inn}K$ nonabelian, so the cokernel edge (I) leaves the abelian-kernel world and the butterfly's waist class becomes a genuinely nonabelian datum — the Breen bitorsor register [2] that the Aldrovandi–Noohi formalism was built to carry [1]. Neither is claimed here; both are *open*, and the butterfly picture is what makes them precise.

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